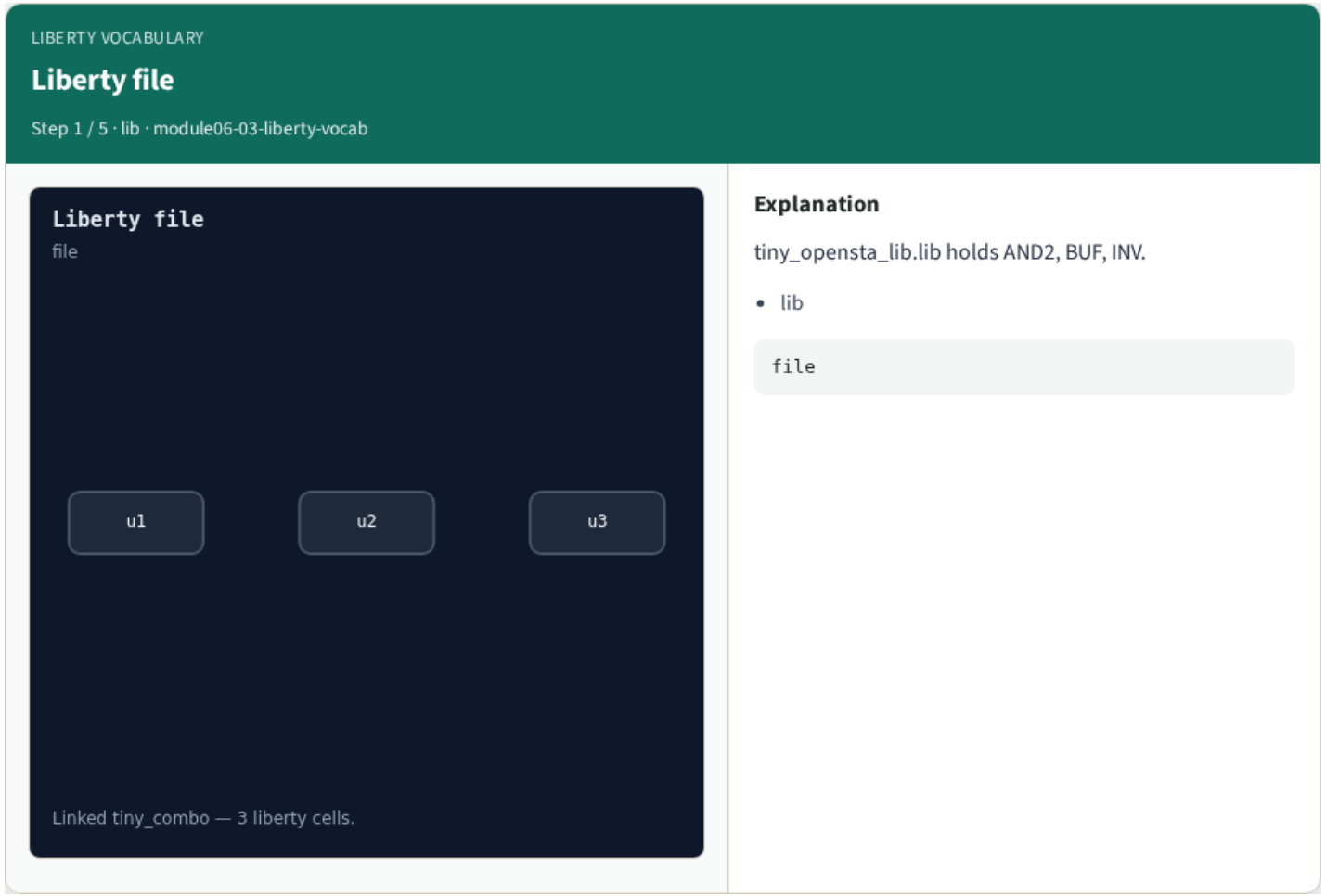
Why real OpenSTA

Name liberty cells AND2/BUF/INV used by tiny_combo

The idea

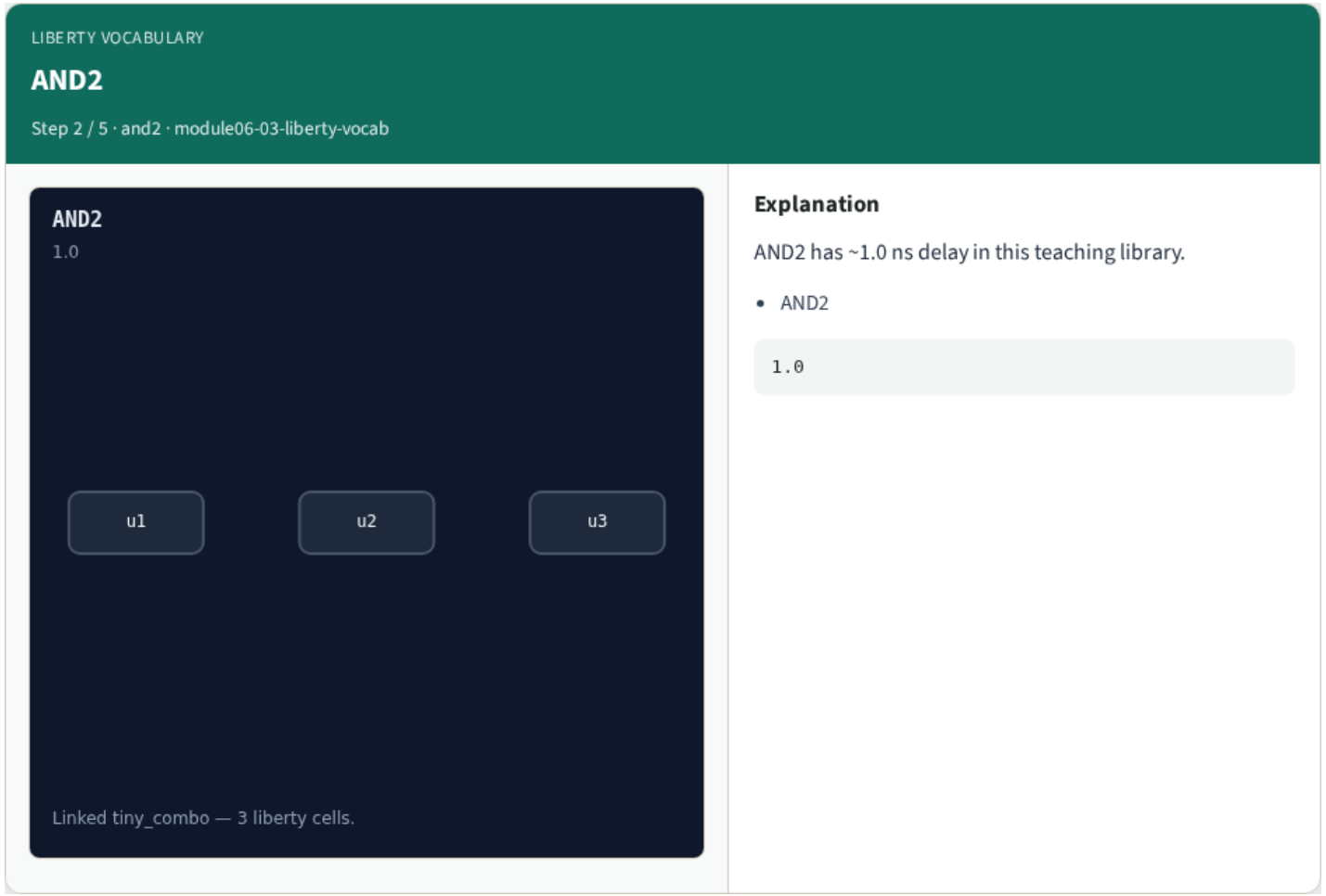
- GOLDEN: Liberty cells AND2, BUF, INV; linked cells=3

Liberty file



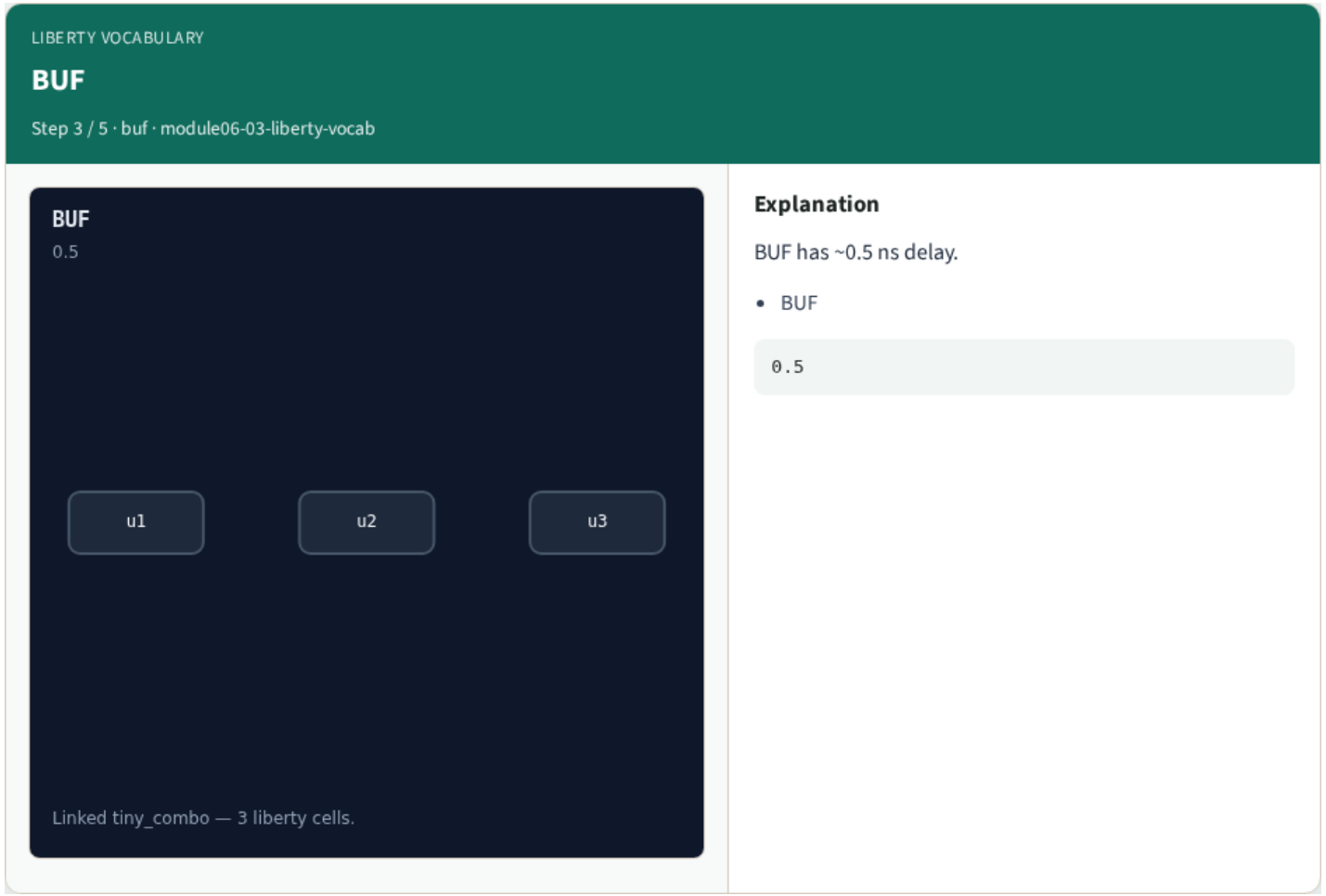
Liberty file

AND2


LIBERTY VOCABULARY
AND2
Step 2 / 5 · and2 · module06-03-liberty-vocab
AND2
1.0
u1
u2
u3
Linked tiny_combo — 3 liberty cells.
Explanation
AND2 has ~1.0 ns delay in this teaching library.
AND2
1.0

AND2

BUF


LIBERTY VOCABULARY
BUF
Step 3 / 5 · buf · module06-03-liberty-vocab
BUF
0.5
u1
u2
u3
Linked tiny_combo — 3 liberty cells.
Explanation
BUF has ~0.5 ns delay.
BUF
0.5

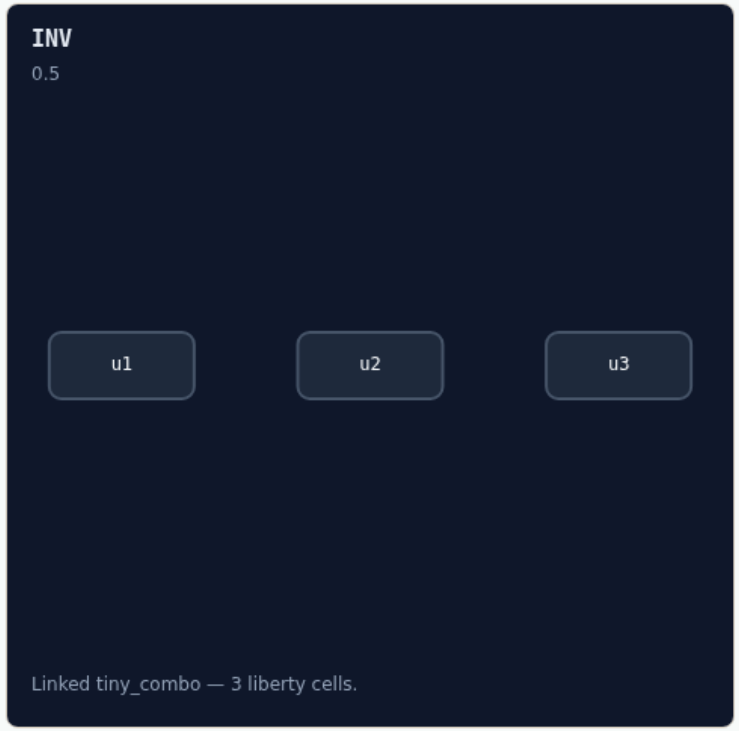
BUF

INV

LIBERTY VOCABULARY

INV

Step 4 / 5 · inv · module06-03-liberty-vocab



Explanation

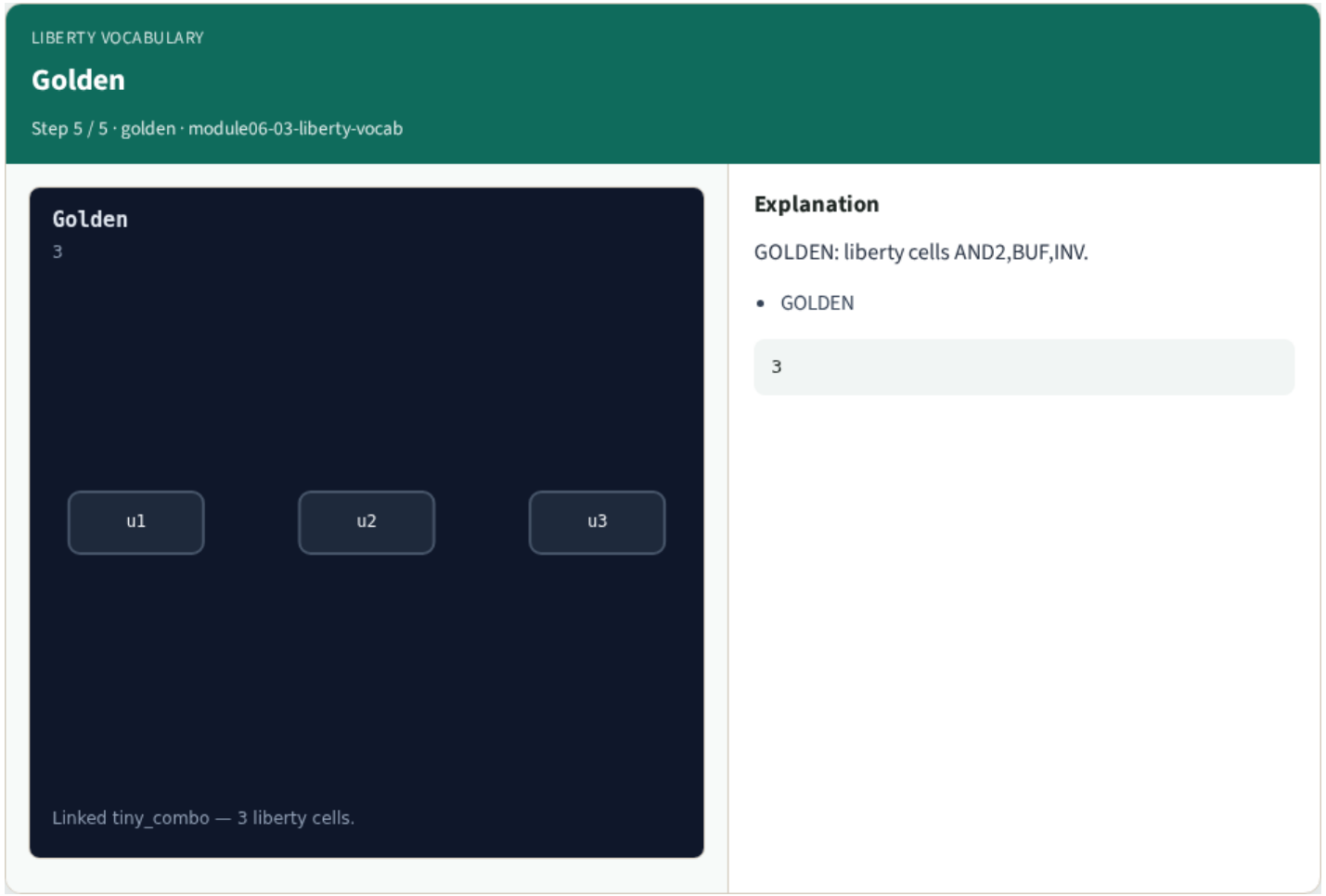
INV has ~0.5 ns delay.

- INV

```
0.5
```

INV

Golden



Golden

Browser lab track

- Open **liberty-vocab**
- Build the Tcl recipe or open evidence stages, then Check

Implement track

- From common/, run real_solvers.py and confirm REAL_OPENSTA_OK

Pitfalls

- Editing goldens by hand without re-running sta
- Confusing Part A map JSON with real liberty delays

Try these

- From the common folder, run the real OpenSTA self-test

Try these — try these

```
cd courses/learn_opensta/common
python3 real_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
Liberty cells AND2, BUF, INV; linked cells=3.
ok if goldens match
```

Next

- Complete the quiz for this part
- Continue the Part B evidence track, then wrap , algorithms and internals come later